1. (a) Simplify $\sqrt[3]{24} + \sqrt[3]{81} + \sqrt{\sqrt[3]{9}}$. [3]

   (b) Rationalise the denominator in the expression $\dfrac{1}{\sqrt[3]{x} - 1}$. [4]

2. (a) Express the set $\{x | x \neq \pm 1\}$ of real numbers using interval notation. [3]

   (b) Evaluate $|x - 2| + |x - 3|$ given that $x < 2$. [6]

3. (a) Determine whether or not the graph of $y = 2x^2 - x^3 - x^5 + x^7$ is symmetrical about the y-axis, the x-axis or the origin. [6]

   (b) Find the value(s) of k such that the equation $3x^2 + (\sqrt{2k})x + 6 = 0$ has exactly one root. [6]

4. Solve the following equation for x

   (a) $x^{4/3} + 3x^{2/3} - 28 = 0$. [6]

   (b) $4 + f(x + 3) = -3; f^{-1}(-7) = 0$, by assuming that the domain for f and f^{-1} is $(-\infty, \infty)$. [6]

5. Solve the following inequality. [8]

$$\left|\frac{x - 2}{3}\right| - 2 \leq 2.$$

6. (a) Let $f(x) = \dfrac{x + 2}{x - 3}$. Find the domain and range of the function $f(x)$. [6]

   (b) Let $f(x) = 2^{x-1}$ and $g(x) = x^2 + 1$. Evaluate $(g \circ f)(x)$. [3]

7. Let $f(x) = e^{-x+1}$. Sketch the graph of $f(x)$ and specify intercepts and asymptotes. [6]

8. (a) Solve the equation: $8^{2x+1} = 16\sqrt{2}$. [5]

   (b) Simplify $\log_b \sqrt{\dfrac{x}{b}}$ so that the logarithm of quotients does not appear. [4]

9. Convert

   (a) $330°$ to radians. [2]

   (b) $\dfrac{9\pi}{5}$ radians to degrees. [2]

10. Evaluate each expression.

    (a) $\sec 240°$. [5]

    (b) $\cot\left(-\dfrac{\pi}{6}\right)$. [6]

11. Verify the following identity: [7]

$$\sin A + \cos A = \frac{\sin A}{1 - \cot A} - \frac{\cos A}{\tan A - 1}.$$

END OF QUESTION PAPER

1. (a) Simplify $\left(\frac{1}{4}\right)^{-2} + \left(\frac{2}{3}\right)^{-2}$.

   (b) Evaluate $\sqrt{\sqrt[3]{64}}$.

2. Rationalise the numerator in the expression

$$\frac{\sqrt{x+h} - \sqrt{x}}{h}.$$

3. Rewrite the following expression without using absolute values given that $x < 3$

$$|x^2 + 1| + |x - 3|.$$

4. Determine the center and radius of a circle with the equation $x^2 + y^2 - 10x + 2y + 17 = 0$.

5. Solve the inequality and specify the answer using interval notation.

$$\frac{x+3}{x-4} \geq 0$$

6. Let $f(x) = \dfrac{x+2}{x-3}$.

   (a) Find the domain and range of the function $f(x)$.

   (b) Find $f^{-1}(x)$.

7. Use translations and reflections to sketch the graph of $f(x) = -e^x + 1$. Also specify the domain, range, intercepts and asymptotes of $f(x)$.

8. Show that $\log_b\left(\dfrac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}\right) = 2\log_b(\sqrt{3}+\sqrt{2})$ by using an appropriate rationalization.

9. Solve $3(\ln x)^2 - \ln x^2 - 8 = 0$ for x.

10. Convert

    (a) $450°$ to radians.

    (b) $\dfrac{11\pi}{6}$ to degrees.

11. (a) If $\sec\theta = \frac{7}{5}$ and $270° < \theta < 360°$, find $\tan\theta$.

    (b) Evaluate $\cos\left(-\dfrac{5\pi}{4}\right)$.

12. (a) Simplify the expression $\dfrac{\cot^2\theta}{\csc^2\theta} + \dfrac{\tan^2\theta}{\sec^2\theta}$.

    (b) Prove the identity $\sin A + \cos A = \dfrac{\sin A}{1 - \cot A} - \dfrac{\cos A}{\tan A - 1}$.

1. (a) Rationalise the numerator in the expression $\dfrac{2\sqrt{y}+3}{4y-9}$ and give your answer in the simplest form. [3]

   (b) Factorise and simplify the expression $A^3 + B^3 + 3AB(A+B)$ completely. [4]

2. (a) Let $A = \{x \mid x \in (-\infty, 2]\}$ and $B = \{x \mid x \in (1, \infty)\}$. Compute $A \cap B$. [3]

   (b) Evaluate $|x-3| + |x-4|$ given that $x < 3$. [6]

3. (a) Determine whether or not the graph of $y = 2x - x^3 - x^5 + x^7$ is symmetrical about the y-axis, the x-axis or the origin. [6]

   (b) Find the value(s) of k such that the equation $3x^2 + (\sqrt{2k})x + 6 = 0$ has exactly one root. [6]

4. Solve the following equations.

   (a) $x^{4/3} + 3x^{2/3} - 28 = 0$. [6]

   (b) $\sqrt{x-5} - \sqrt{x+4} + 1 = 0$. [6]

5. Solve the following inequality. [8]

$$\left|\frac{x+1}{2} - \frac{x-1}{3}\right| < 1.$$

6. (a) Let $f(x) = \dfrac{2}{x}$. Find $\dfrac{f(x+h) - f(x)}{h}$ and simplify the expression. [5]

   (b) Let $f(x) = 2^x$ and $g(x) = x^2 + 1$. Evaluate $(f \circ g)(x)$. [3]

   (c) Given that $f(x) = \dfrac{x}{3x-6}$, find $f^{-1}(x)$ and determine its range. [6]

7. Let $f(x) = -e^{x+1}$. Sketch the graph of $f(x)$ and specify intercepts and asymptotes. [6]

8. (a) Solve the equation: $8^{x+1} = 32\sqrt{2}$. [5]

   (b) Given that $xy = 8$, evaluate $5\log_2 x + 5\log_2 y$. [5]

9. Convert

   (a) $450°$ to radians. [2]

   (b) $\dfrac{11\pi}{6}$ radians to degrees. [2]

10. Evaluate each expression.

    (a) $\csc 240°$. [5]

    (b) $\tan\left(-\dfrac{\pi}{3}\right)$. [6]

11. Verify the following identity: [7]

$$\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} = 2\csc\theta.$$

END OF QUESTION PAPER

1. (a) Factorise $x(2x-1)^{-\frac{1}{2}} + (2x-1)^{\frac{3}{2}}$.

(b) Simplify $\dfrac{\frac{4}{a} - a}{1 + \frac{2}{a}}$.

(c) Simplify $\sqrt{2} - \sqrt{12} + \sqrt{48}$

2. Evaluate

(a) $\sec(-765°)$

(b) $\csc \frac{4\pi}{3}$.

3. Write as a rational exponent

$\sqrt{\sqrt[3]{x}\sqrt[4]{y}}$

4. (a) Let $\log_b 2 = A, \log_b 3 = B$ and $\log_b 5 = C$. Evaluate $\log_b 0.6$ in terms of A, B and C. [4]

(b) Solve the equation $3x(10^x) + 10^x = 0$. [3]

(c) Sketch the graph $y = -\ln(x-2)$, and specify the domain , range, asymptote and intercept. [6]

5. Let $f(x) = \frac{x-1}{3x+5}$.

(a) Find $f(m-2)$. [2]

(b) Find $f^{-1}(x)$. [4]

6. Let $g(x) = \frac{1}{x}$. Find $\dfrac{g(x+h) - g(x)}{h}$.

7. Determine the center and the radius for the circle $x^2 + y^2 + 8x - 6y = -24$.

8. Rationalise the denominator

   (a) $\dfrac{\sqrt{2}}{1 - \sqrt{2}}$

   (b) $\dfrac{1}{\sqrt[3]{a} - 1}$

9. Solve the inequality [8]

$$\frac{x+1}{x+2} > \frac{x-3}{x+4}.$$

10. (a) Prove the identity $(\sec\alpha - \tan\alpha)^2 = \frac{1-\sin\alpha}{1+\sin\alpha}$. [5]

(b) Let $y = 2\sin 2\theta$.

i. Find amplitude and period for y. [3]

ii. Sketch the graph of y. [6]

MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS AND STATISTICS

MATH 1201 - CALCULUS

END OF SEMESTER EXAMINATIONS

Date: 11 March, 2019 Time Allowed: 3 hours (8:00-11:00 AM)

INSTRUCTIONS

- Write your name, registration number, program and level in the space provided below.
- Check that the paper has **12** questions.
- Attempt **all** questions.
- Write your answer in the space provided under each question.
- Show all your working.
- Calculators are not allowed.

Name :

Registration Number:

Program: :

Level: :

1. Evaluate

(a) $\lim_{x \to 1} \dfrac{x^2 - 1}{x^2 - 3x + 2}$,

(b) $\lim_{x \to \infty} \dfrac{3x^2 - 6x + 2}{2x - 9}$.

2. Determine if the function is continuous or not.

$$f(x) = \begin{cases} 5 - x & \text{if } x \leq 2 \\ 5x - 3 & \text{if } x > 2 \end{cases}.$$

3. Find the derivative of $f(x) = \frac{1}{\sqrt{x}}$ using the definition of derivative (First principles). [5]

4. Differentiate

(a) $y = \sin\left(e^{x^2}\right)$, [3]

(b) $y = \dfrac{3x-1}{2x+1}$. [5]

5. Use implicit differentiation to find $\frac{dy}{dx}$ if $x^2 - xy = 5$.

6. Find equation of the tangent line to the curve $x^2 + 3x + 1$ at $x = 0$.

7. Let $f(x) = x^3 - 3x + 3$.

(a) Find critical numbers. [4]

(b) Determine the intervals where the function is

i. increasing. [2]

ii. decreasing. [2]

(c) Sketch the graph. [5]

8. Use Riemann sum to evaluate $\int_0^3 (x^2+1)\,dx$ using right endpoint with $n = 6$.

9. Use the Fundamental Theorem of Calculus Part 1 to find the derivatives of $F(x) = \int_{-1}^{x} \sqrt{t^4+1}\,dt$.

10. Evaluate the integrals

(a) $\int_0^2 2x(x^2+1)\,dx$, [4]

(b) $\int \sin 2x \cos 2x\,dx$. [4]

11. Use partial fractions to evaluate $\int \frac{x^2 + 12x + 12}{x^3 - 4x}\, dx$.

12. Find the volume of the solid formed by revolving the region between $y = x^2$ and $y = x^3$ about y-axis. [7]

END OF QUESTIONS

MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS AND STATISTICS

MATH 1101- PRECALCULUS

End of Semester Examinations

Date: Monday, 12 August, 2019 Time Allowed: 3 hours (8:00-11:00 AM)

INSTRUCTIONS

- Write your registration number and lecturer's name in the spaces provided below.
- Check that the paper has **11** questions.
- Attempt **all** questions.
- Write your answer in the space provided under each question
- Show all your working.
- Calculators are not allowed.

Registration Number: _____

Lecturer's Name : _____

Question:	1	2	3	4	5	6	7	8	9	10	11	Total
Points:	7	5	10	13	8	8	6	10	16	10	7	100
Score:												

1. Simplify each of the following expressions.

   (a) $(10^{-1} + 10^{-2})^{-1}$.

   (b) $\sqrt[3]{24} + \sqrt[3]{81} + \sqrt{\sqrt[3]{9}}$.

2. Determine the center and radius of a circle with the equation $x^2+y^2-10x+2y+17 = 0$.

3. (a) Let $f(x) = -2x^2 + 6x + b$ and $f(0) = -1$, find the value of b.

(b) If $g(x) = x^2 + 3x$, find $\dfrac{g(x) - g(2)}{x - 2}$ in simplified form. [4]

(c) If $f(x) = 3x - 1$, find $f^{-1}(x)$. [4]

4. Solve each of the following equation.

(a) $\sqrt{1 - 2x} + \sqrt{x + 5} = 4$. [5]

(b) $|x - 5| = 1$.

(c) $\log_2(x^2 - 4) = 5$.

5. Solve the inequality $x^2 - 3x - 4 \geq 0$.

6. (a) Determine the domain of the function $y = \sqrt{-5x + 1}$. [4]

(b) Determine the range of $y = \sqrt{\dfrac{x+3}{x-4}}$. [4]

7. Without using the table of values, sketch the graph of $y = -e^{x-1} + 1$ specifying the domain, range, intercept, and asymptote. [6]

8. (a) Evaluate the following.

i. $\log_2 8\sqrt{2}$.

ii. $2^{\log_2 5} - 3\log_5 \sqrt[3]{5}$

(b) If $\log_b 2 = A$ and $\log_b 6 = B$, evaluate $\log_b \left(\frac{b}{36}\right)$.

9. (a) i. Convert $150°$ to radians.

ii. Convert $\dfrac{2\pi}{3}$ to degrees.

(b) Evaluate each expression

i. $\sin 600°$.

ii. $\tan\left(-\frac{\pi}{3}\right)$. [4]

iii. Evaluate $\cos\left(\frac{13\pi}{3}\right)$. [4]

10. Prove each of the following identities.

(a) $\dfrac{\sin\theta}{1+\cos\theta} + \dfrac{1+\cos\theta}{\sin\theta} = 2\csc\theta$ [5]

(b) $\tan A \tan B = \dfrac{\tan A + \tan B}{\cot A + \cot B}$.

11. Sketch the graph of $y = 3\cos\left(x - \dfrac{\pi}{2}\right)$, specify the amplitude, period and $x-$intercept

End of Question Paper

MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS AND STATISTICS

EMT 1201 - CALCULUS I

END OF SEMESTER EXAMINATIONS

Date: 18 March, 2019 Time Allowed: 3 hours (8:00-11:00 AM)

INSTRUCTIONS

- Check that the paper has **10** questions.
- Attempt **all** questions.
- Show all your working.
- Calculators are not allowed.

1. Evaluate
   (a) $\lim_{x\to 1} \dfrac{\sqrt{x}-1}{x-1}$,
   (b) $\lim_{x\to\infty} \dfrac{x^2-2x+3}{2x^2+5x+1}$,
   (c) $\lim_{x\to 0} \dfrac{\sin 2x}{\sin 5x}$.
2. If $\lim_{x\to 3} \dfrac{f(x)-8}{x-1} = 10$, find $\lim_{x\to 3} f(x)$.
3. At what value of the constant c is the following function continuous for all real x.
$$f(x) = \begin{cases} cx+5 & \text{if } x < 1 \\ x^2-3x+4 & \text{if } x \geq 1 \end{cases}.$$
4. Find the derivative of $f(x) = 1-\sqrt{x}$ using the definition of derivative (First principles).
5. Differentiate
   (a) $y = \cos^2 x$,
   (b) $y = \sec x + \tan^2 x$,
   (c) $y = \ln(x^2-1)$,
6. Find the second derivative of $f(x) = \sqrt{x^2+1}$.
7. (a) Use implicit differentiation to find $\dfrac{dy}{dx}$ of $x^2y^2 - 2x = 3$.
   (b) Find equation of the tangent line to the curve $xy = 4$ at a point $(-4,-1)$. (*Hint: Use implicit differentiation*).
8. If $F(x) = f(g(x))$ where $f(-2) = 8$, $f'(-2) = 4$, $f(5) = 3$, $g(5) = -2$ and $g'(5) = 6$, find $F'(5)$.
9. Let $f(x) = x^2+2x-4$ be a continuous function on the interval $(-1,1)$.
   (a) Find critical numbers.
   (b) Determine intervals where the function
      i. increasing.
      ii. decreasing.
   (c) Find the
      i. local maximum .
      ii. local minimum.
   (d) Sketch the graph.
10. Find all numbers c that satisfy the Mean Value Theorem given that $f(x) = 3x^2+2x+5$ on interval $[-1,1]$.

MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS

MATH 1101- PRECALCULUS

End of Semester Examination

Date: Wednesday, 14th February, 2018 Time Allowed: 3 hours (08:00-11:00 AM)

NSTRUCTIONS

- Write your registration number, programme and lecturer's name in the spaces provided below
- Check that the paper has **12** questions
- Attempt **all** questions
- Write your answer in the space provided under each question
- Show all your working
- Calculators are not allowed

egistration Number:

rogramme:

cturer's Name :

uestion:	1	2	3	4	5	6	7	8	9	10	11	12	Total
oints:	12	9	9	7	11	11	6	10	4	7	7	7	100
core:													

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1. (a) Factorise $x(2x-1)^{-\frac{1}{2}} + (2x-1)^{\frac{3}{2}}$.

(b) Find all real value solutions for the equation, $9x^{\frac{4}{3}} - 10x^{\frac{2}{3}} + 1 = 0$.

2. (a) Evaluate $\sqrt[4]{\frac{64a^{\frac{2}{3}}}{a^{\frac{1}{3}}}}$.

(b) Rationalise the denominator in $\frac{1}{\sqrt[3]{a}-1}$.

3. Solve the inequality $\frac{x+1}{x+2} > \frac{x-3}{x+4}$. [9]

4. Determine the domain and range of the function $g(x) = \frac{4x-20}{3x-18}$ and present your answer in interval notation. [7]

5. (a) Solve for x, $7^{-4x} = 2^{1+3x}$. [6]

(b) Sketch the graph of the function, $y = |4 - x|$ without using table of values. [5]

6. Evaluate each expression.

   (a) $\csc \dfrac{4\pi}{3}$

   (b) $\sin 105°$.

7. Prove the equation $(1 - \sin\theta)(\sec\theta + \tan\theta) = \cos\theta$.

8. (a) Let $g(x) = 2x + 1$. Find $f(x)$, given that $g(f(x)) = 10x - 7$.

(b) Let $f(x) = 2^x$. Find $\dfrac{f(x+h) - f(x)}{h}$ and show that it reduces to $2^x\left(\dfrac{2^h - 1}{h}\right)$. [5]

9. Let $\log_b 2 = A, \log_b 3 = B$ and $\log_b 5 = C$. Evaluate $\log_b 0.6$ in terms of A, B and or C. [4]

10. Suppose that β is an acute angle and $\sin\beta = \dfrac{m^2 - n^2}{m^2 + n^2}$ $(m > n > 0)$. Show that $\cos\beta = \dfrac{2mn}{m^2 + n^2}$. [7]

11. Sketch the graph $f(x) = -cos\left(x + \frac{\pi}{2}\right)$, $-2\pi < x < 2\pi$. [7]

12. Solve the equation, $2\sin^2 x = 3\sin x - 1, 0 \leq x \leq 2\pi$ [7]

END OF QUESTIONS

MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS AND STATISTICS

MATH 1101 – PRECALCULUS

FINAL EXAM

Date: February 11, 2019 Time Allowed: 3 hours (14:00 – 17:00 HRS)

INSTRUCTIONS

- Write your registration number, programme and lecturer's name in the spaces provided below
- Check that the paper has **12** questions
- Attempt **all** questions
- Write your answer in the space provided under each question
- Show all your working
- Calculators are not allowed

Registration Number: _____

Programme: _____

Lecturer's Name: _____

Question:	1	2	3	4	5	6	7	8	9	10	11	12	Total
Points:	6	6	8	8	11	5	6	7	10	15	7	11	100
Score:													

-1. Reduce the following fraction to its lowest form.

$$\frac{\left(a+\frac{1}{b}\right)^a\left(a-\frac{1}{b}\right)^b}{\left(b+\frac{1}{a}\right)^a\left(b-\frac{1}{a}\right)^b}.$$

2. Find the value of k such that the following equation has exactly one root.

$$3x^2+(\sqrt{2k})x+6=0.$$

5. Express the domain of the following functions in interval notation.

(a) $f(x) = \dfrac{4}{2 - \sqrt{x+3}}$. [6]

(b) $h(x) = \log_{10} \dfrac{2x+3}{x-5}$. [5]

6. Let $f(x) = 2x^3 + 1$. Find $f^{-1}(x)$. [5]

3. Solve the following inequality and put your answer in interval notation.

$$\left|\frac{x+1}{2} - \frac{x-1}{3}\right| < 1.$$

4. Solve the equation $\sqrt{x+2} + \sqrt{x-2} = \sqrt{4x-2}$.

(b) Solve for x: $\log_2(\log_3 x) = -1$.

10. Evaluate each expression.

(a) $\csc \dfrac{4\pi}{3}$.

(b) $\tan\left(-\dfrac{\pi}{3}\right)$.

(c) $\cos \dfrac{13\pi}{3}$.

7. Let $f(x) = -2x+1$ and $g(x) = ax+b$. Find a and b so that the equation $(f \circ g)(x) = x$ holds for all values of x. [6]

8. Consider the function $y = 2x^2+3x-2$. Find the vertex, minimum or maximum value and intercept(s) of its graph. [7]

9. (a) Simplify $2^{\log_2 5} - 3\log_5 \sqrt[3]{5}$. [4]

11. Verify the identity [7]

$$\frac{\sin\theta}{1+\cos\theta}+\frac{1+\cos\theta}{\sin\theta}=2\csc\theta.$$

12. Sketch the graph the following function on the interval $-\pi \leq \theta < 2\pi$. Specifying the amplitude, period, x-intercept, and intervals on which the graph is increasing. [11]

$$y = 2\sin\left(x-\frac{\pi}{2}\right).$$

END OF QUESTION PAPER

MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS AND STATISTICS

END OF SEMESTER EXAMINATIONS

MATH 1101 - PRECALCULUS

Friday, March 8, 2019 TIME: 2:00 – 5:00 PM (3 Hours)

INSTRUCTIONS

1. Attempt all questions.
2. All working must be clearly shown.
3. Check that the paper has 8 questions.

1. a. Express each interval below using inequality notation and show the given interval on a number line.

   i. $(\sqrt{2}, \infty)$ (2 marks)

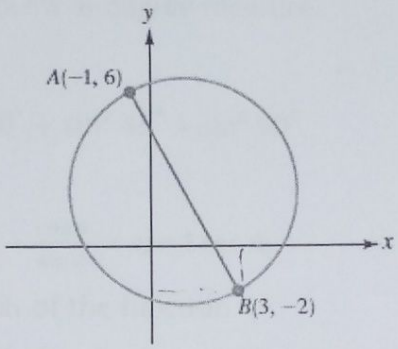
   ii. $(-\infty, \pi]$ (2 marks)

   b. Simplify $|x-3| + |x-4|$ given that $3 < x < 4$. (3 marks)

   c. The distance between y and -4 is less than 1. Rewrite the statement using absolute value notation. (1 mark)

   d. Given that $a = -2$, $b = 3$ and $c = -4$, evaluate $|a+b|^2 - |b+c|^2$ (4 marks)

2. The points A (-1, 6) and B (3, -2) are the endpoints of a diameter of a circle, as indicated in figure below.



   Find:

   a) The equation of the circle. (8 marks)

   b) The y-intercepts of the circle. (4 marks)

3. Solve the following inequalities, state the solution in interval notation and graph the solution set.

   a. $\left|1 - \frac{t}{2}\right| > 5$ (7 marks)

   b. $x^3 + 3x^2 > 4x$ (11 marks)

4. Test the equation $y = x^3 + 5x$ for symmetry about

   a) the x-axis. (2 marks)

   b) the y-axis. (2 marks)

   c) the origin. (2 marks)

5. Consider the function $y = \sqrt{\frac{x-2}{2x+6}}$. Find

   i. its domain, (8 marks)

   ii. its range. (7 marks)

6. Let $f(x) = 2x + 3$. Find

   a. $f(ax + b)$, (2 marks)

   b. a and b, if $f(ax + b) = x$. (4 marks)

7. a. Express 144° in radian measure as a multiple of π. (2 marks)

   b. Express $\frac{11}{6}\pi$ radians in degree measure. (2 marks)

   c. Evaluate $\sin^2 30^\circ + \sin^2 45^\circ + \sin^2 60^\circ$. (5 marks)

   d. Show that $\frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} = \sec A \csc A$. (6 marks)

8. a. Sketch the graph of the function

$$C(x) = \begin{cases} x^3, & if\ x < 1 \\ \sqrt{x}, & if\ x > 1 \end{cases}$$ (3 marks)

   b. Compute $(f \circ g)(x)$ for the pair of functions $f(x) = 2^x$ and $g(x) = x^2 + 1$. (3 marks)

END OF QUESTIONS

MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS AND STATISTICS

MATH 1101 – PRECALCULUS

SUPPLEMENTARY/DEFERRED EXAM

Date: Friday, May 10, 2019 Time Allowed: 3 hours (08:00 – 11:00 AM)

NSTRUCTIONS

- Write your registration number, programme and lecturer's name in the spaces provided below
- Check that the paper has **12** questions
- Attempt **all** questions
- Write your answer in the space provided under each question
- Show all your working
- Calculators are not allowed

egistration Number: _____

ogramme: _____

cturer's Name: _____

uestion:	1	2	3	4	5	6	7	8	9	10	11	12	Total
oints:	7	8	8	8	6	7	6	7	9	16	8	10	100
core:													

1. Factorise each of the following expression completely.

$$A^3 + B^3 + 3AB(A + B).$$

2. By using the quadratic formula solve the following equation.

$$x^2 - (\sqrt{2} + \sqrt{5})x + \sqrt{10} = 0.$$

Solve the following inequality and leave the final answer in interval notation. [8]

$$\frac{3x+1}{x-4} < 1.$$

Solve the equation $\sqrt{x+2} + \sqrt{x-2} = \sqrt{4x-2}$. [8]

5. Express the domain of the following functions in interval notation.

   (a) $g(x) = \sqrt{4 - 3x}$.

   (b) $h(x) = \sqrt[5]{4 - 3x}$.

6. By showing all the asymptotes and intercepts clearly. Sketch the graph of $y = 1 + \dfrac{-2}{x+1}$.

7. Let $f(x) = \dfrac{3x-4}{3x+3}$ and $g(x) = x - 1$. Then evaluate $(f \circ g)(x)$, leaving the answer its simplest form.

8. Consider the graph of $y = 2x^2 + 3x - 2$. Find the vertex, minimum or maximum value and intercept(s). [7]

9. (a) Simplify $e^{\ln 3} + e^{e \ln 2} - e^{2 \ln e}$. [4]

(b) Solve for x: $\log(x^2 - 3x) = 1$. [5]

0. Evaluate the following.

(a) $\cos(-150°)$. [6]